

FUNDAMENTALS · TRANSPORT LAYER · DEEP DIVE NOTES

Module 2: TCP vs UDP

Guaranteed In-Order Byte Streams vs Zero-Overhead Real-Time Datagrams

Author & Educator Notes · Submodule 02 · Target Audience: CS / Web Engineering Students

The Core Tradeoff in Networking: You cannot have absolute 100% reliable delivery and zero latency simultaneously over an unreliable physical medium. Every distributed system must make a choice: *Do you wait for missing bytes (TCP), or do you drop them and keep moving forward in real time (UDP)?*

1 Why IP Is Not Enough: The Birth of Transport Layer & Ports

THE LIMITATION OF INTERNET PROTOCOL (LAYER 3)

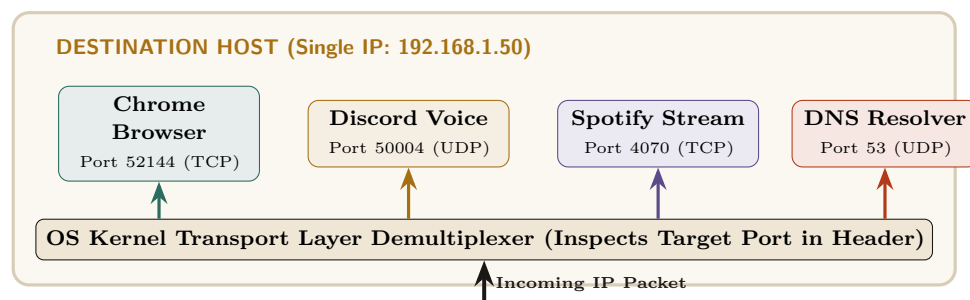
IP (Internet Protocol) solves **Host-to-Host delivery**. It routes a packet from an IP in Tokyo to an IP in London. But here is the fatal limitation:

- A computer does not run just one program. A typical developer's laptop has **Chrome (15 tabs)**, **VS Code**, **Discord**, **Spotify**, and a **background database** running simultaneously.
- When a packet arrives at your machine's IP, which program should receive it?
- IP is completely **unreliable**: packets can drop, arrive out-of-order, duplicate, or be corrupted by noisy switches.

THE SOLUTION: PROCESS-TO-PROCESS ADDRESSING VIA PORTS

Layer 4 (Transport Layer) introduces **Port Numbers (0 to 65,535)**.

- **IP Address** = The Apartment Building Address (Host).
- **Port Number** = The Specific Flat Number / Room inside the building (Process/Application).
- A complete network socket is identified by a 4-tuple: (Source IP, Source Port, Dest IP, Dest Port).



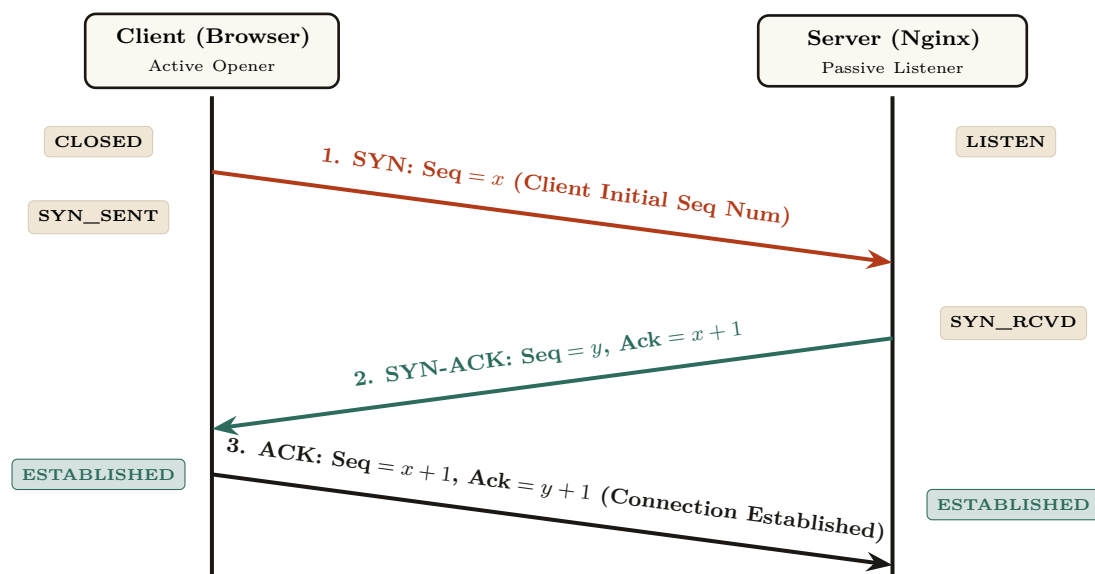
2 TCP (Transmission Control Protocol): The Reliable Contract

TCP FUNDAMENTALS (RFC 793 / RFC 9293)

NATURE	Connection-oriented, full-duplex, reliable, point-to-point byte-stream protocol.
KEY GUARANTEES	1. Zero packet loss (via Retransmission) 2. Strict in-order arrival (via Sequence Numbers) 3. Zero buffer overflow (Flow Control) 4. Network collapse prevention (Congestion Control).
HEADER SIZE	20 to 60 bytes (minimum 20 bytes when no options are set).

2.1 The 3-Way Handshake (Connection Establishment)

Before transmitting a single byte of user data, TCP establishes state and synchronizes initial sequence numbers (ISN) between client and server.



2.2 The 4 Core Pillars of TCP Reliability

- Sequence & Acknowledgement Numbers:** Every transmitted byte has a distinct index. If bytes 1000–1499 are sent, the receiver responds with $\text{ACK} = 1500$ (“I have received everything up to 1499, now send byte 1500”).
- Retransmission Timer (RTO):** TCP continuously measures Round-Trip Time (RTT). If an ACK is not received within RTO, TCP automatically retransmits the segment.
- Sliding Window Flow Control (Receiver Protection):** The receiver includes a **Window Size** (*wnd*) in every header, telling the sender: “My RAM buffer has only 64 KB free right now; do not send more than 64 KB without waiting for my next ACK.” This prevents a fast sender from crashing a slow receiver.

3 UDP (User Datagram Protocol): The Speed Demon

UDP FUNDAMENTALS (RFC 768)

NATURE	Connectionless, stateless, minimal, message-oriented datagram protocol.
GUARANTEES	None. No handshake, no delivery confirmation, no ordering, no retransmission, no flow control, no congestion control.
HEADER SIZE	Fixed 8 bytes (Source Port, Destination Port, Length, Checksum).
CORE MOTTO	<i>“Deliver fast if possible; if dropped, do not look back and do not apologize.”</i>

TCP HEADER (MINIMUM 20 BYTES)

Source Port (16 bit)	Dest Port (16 bit)
Sequence Number (32 bit)	
Acknowledgement Number (32 bit)	
Offset (4b) Flags (SYN/ACK/FIN/RST) Window (16b)	
Checksum (16b)	Urgent Ptr (16b)

UDP HEADER (EXACTLY 8 BYTES)

Source Port (16 bit)	Dest Port (16 bit)
Length (16 bit)	Checksum (16 bit)

60% Less Header Overhead!

Zero connection state stored in OS kernel.

■ WHY WOULD ANYONE WANT UDP?

“Agar ek student puche: ‘Sir jab TCP sab guarantee deta hai toh koi pagal hi hoga jo UDP use karega?’

Aap unhe counter-question karein: ‘Socho aap Zoom call par bol rahe ho: **“Aap meri aawaz sun sakte ho?”**’

Agar word **“meri”** ka packet raaste mein drop ho jaye:

- **TCP:** Poora audio freeze ho jayega, TCP missing packet ko dobara maangega, 1 second baad aawaz aayegi: “Aap... (silence)... meri aawaz sun sakte ho?”
- **UDP:** Ek syllable halka sa distort hoga, lekin realtime conversation chalti rahegi! **Realtime communication mein late data is equal to garbage data.**”

4 Head-of-Line (HoL) Blocking & Use Case Selection

THE INHERENT FLAW OF TCP: HEAD-OF-LINE (HOL) BLOCKING

Because TCP guarantees strict ordering, if Packet 1 arrives, **Packet 2 is dropped**, and Packets 3, 4, 5 arrive safely — the operating system **refuses to give Packets 3, 4, 5 to the application**. They sit idle in the kernel buffer until Packet 2 is re-requested, re-sent, and acknowledged.



TCP KERNEL LOCK: Packets 3, 4, 5 CANNOT be delivered to the application until Packet 2 is retransmitted!

(Because TCP enforces in-order byte streams, 1 dropped packet stalls all subsequent received packets in memory).

4.1 When to Use Which? The Engineering Rulebook

Domain	Winning Protocol	Why This Choice Is Mandatory
Web Pages (HTTP/1.1 & HTTP/2)	TCP	Missing a few HTML/JS bytes corrupts the entire web page logic.
File Transfer (FTP, SFTP)	TCP	A single corrupted bit in an executable file renders it unrunnable.
Emails (SMTP, IMAP)	TCP	Message content must arrive 100% complete and uncorrupted.
Live Multiplayer Gaming	UDP	Game state updates 60 times/sec; a 200ms-late position update is obsolete.
Live Video / Voice (WebRTC, Zoom)	UDP	Human ear tolerates tiny millisecond drops, but cannot tolerate latency lag.
DNS Queries	UDP	Single request, single response. Handshake overhead would double lookup time.
Modern Web (HTTP/3 & QUIC)	UDP + Custom User-Space Logic	Solves TCP Head-of-Line blocking while implementing custom encryption & streams.

5 Master Comparison Matrix: TCP vs UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	Connection-oriented (Handshake required)	Connectionless (Send datagram directly)
Reliability	Guaranteed (Retransmits lost packets)	Unreliable (Best-effort; drops allowed)
Packet Ordering	Guaranteed strict sequential ordering	Packets may arrive out-of-order or duplicate
Header Size	20 to 60 bytes	Exactly 8 bytes
Speed & Latency	Slower (Handshake + ACK roundtrips)	Blazing fast (Zero connection latency)
Data Flow Style	Continuous Byte Stream (No message boundaries)	Discrete Datagrams (Preserves packet boundaries)
Flow Control	Yes (Sliding window prevents receiver overflow)	No (Receiver can be overwhelmed)
Congestion Control	Yes (Slow start, AIMD prevents network collapse)	No (Blasts packets at wire speed)
Broadcasting	Unicast only (1-to-1)	Supports Unicast, Multicast, and Broadcast
Key RFC Standard	RFC 793 / RFC 9293	RFC 768

6 Recap: Blackboard Summary & The One-Line Spine

The One-Line Spine of Transport Layer Protocols

CORE TAKEAWAY CHAIN FOR THE BOARD

IP routes host-to-host, but apps need process-to-process → Ports (0-65535)

Need 100% Accuracy & Zero Data Loss? → TCP: 3-Way Handshake + ACKs + Sliding Window

Need Real-Time Speed & Low Latency? → UDP: 8-Byte Header + Fire & Forget

■ CLASSROOM ANALOGY FOR THE FINAL EXAM

“Dimaag mein yeh do tasveerein fit kar lo:

TCP = Registered Post / Courier. Delivery se pehle phone aayega, sign karna padega, receipt milegi. Agar courier kho gaya toh agency naya package bhejegi.

UDP = Loudspeaker Announcement. Jo sun liya so sun liya. Agar truck ke horn ki wajah se do shabd miss ho gaye, toh speaker dobara peechhe jaakar nahi boleگا!”